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D6 / DELIVERY · VERSION 2.0

Implementation Plan

Task dependencies, acceptance tests and release gates

Specification handover · Implementation unverified

Authors

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Issued 11 October 2026 · Asia/Dhaka (UTC+6)

Invented synthetic-data prototype · Interactive team edition

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> Click a section or use the PDF bookmarks. Requirement IDs link to their definitions. External sources are clickable. The meeting workbook contains blank fillable review fields. Application acceptance and pilot evidence remain pending.

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D6 / DELIVERY / V2.0

Implementation Plan

Task dependencies, acceptance tests and release gates · Authors: Sadman Shaharier and Obidit Islam

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Field	Value
Version	2.0
Date	11 Oct 2026 (Asia/Dhaka, UTC+6)
Owner	Sadman Shaharier
Authors	Sadman Shaharier; Obidit Islam
Status	V2 team handover; implementation status UNVERIFIED
Inputs	D1–D5 V2.0; validation report; verified progress and availability still pending.
Key dates	Round 2 submission 14 Oct 2026; exact cutoff/channel pending official email. Mentor dates require team confirmation. Finale 24 Oct 2026 at AIUB.

V2 release contract

Authors: Sadman Shaharier and Obidit Islam. **Version:** 2.0. **Issued:** 11 October 2026, Asia/Dhaka (UTC+6).

This is the revised team specification, not evidence that the application has been built. Actual repository progress, account configuration, interviews, evaluations, and spend remain **UNVERIFIED** until evidence is attached. V1 defaults are superseded by the V2 rules below; existing requirement IDs are retained.

- This release is a **synthetic-data prototype**. Officer verification and referral delivery are simulated and labelled. The schema rejects records marked non-synthetic; the application must also prevent real reports from entering this release.
- A candidate score is similarity, not identity probability. A reviewer decision does not establish guardian authority or safe reunification. A referral row means **referral recorded**, not delivered or acknowledged.
- Draft payload stays on the device before report confirmation, but providers process audio/text beforehand with a notice. Only expiry/counter/hash metadata and cost reservations are stored server-side before submission.
- Possible duplicates remain visible. Shared phone alone never establishes one child. No physical child-return action is performed by Fera.
- Counts distinguish missing-report rows, found-report rows, and confirmed pairs. They are not a census of unique children or proof of rescue.

- Sources, diagrams, tables, code blocks, and acceptance criteria are part of the handover. The companion code must match the embedded code exactly. Team handover records change closure, evidence gaps, and release gates.

Changes to earlier documents

V2 rebases gates from actual remaining work, corrects workload/dependencies, adds TASK-061–064 and TC-038–066, and removes unsupported delivery guarantees. The task estimates and resource totals are computed in the handover.

1 MVP definition

Smallest complete demoable product: one parent, one station volunteer and one officer move one synthetic missing report through a reviewed candidate decision and coordination closure on the map, in Bangla, with a human deciding the match.

The one essential journey

#	Step	Screen	FR
1	Parent records a Bangla voice note	**SCREEN-002**	**FR-002**
2	Fera builds a case draft and asks for the district	**SCREEN-002**	FR-003 – FR-005
3	Parent reviews and submits	**SCREEN-003**	FR-006, FR-007
4	Parent gets the GD draft and poster	**SCREEN-004**	FR-008, FR-009
5	Station volunteer logs a found boy	**SCREEN-007**	**FR-011**
6	Matching surfaces a qualified candidate in the scripted scenario	**SCREEN-009**	FR-012 – FR-014
7	Officer compares evidence and confirms; simulated referral record created	**SCREEN-010**	FR-015, FR-016
8	Officer closes the case; the Dhaka marker updates	SCREEN-012, SCREEN-011	FR-018, FR-019

Included (Must): FR-001 – FR-009, FR-011 – FR-022, FR-025 – FR-028, FR-031 – FR-035, FR-037.

Excluded from MVP: FR-010 13219 summary (Should), FR-024 import (Should), FR-029 edit after submit (Should), FR-023 duplicate screen, FR-030 parent status (Should), FR-036 disaster mode (Could). Alerts, face matching, telecom targeting and live police data are Won't; shown only as [MOCK] on slides.

2 Goals

Type	Goal	Measure	Due
Product	Essential journey works on the deployed app	TC-001 – TC-012 pass	14 Oct 18:00
Product	Safeguarding rules enforced	TC-010, TC-019, TC-022 – TC-027 pass	14 Oct 18:00
Product	Report time beats manual form	Median ≤ 5 min and below manual (GOAL-001)	13 Oct
Technical	Extraction accuracy measured	≥ 85 % target; actual reported (GOAL-002)	13 Oct
Technical	Matching measured	Recall@5 ≥ 80 %, P@1 ≥ 60 % targets; actual reported (GOAL-003)	13 Oct
Technical	Dedupe and WER measured	F1 ≥ 0.8; WER ≤ 30 % targets; actual reported (GOAL-004, GOAL-005)	13 Oct
Technical	AI spend	≤ USD 25 at freeze, ≤ USD 30 at finale (NFR-016)	14 Oct, 24 Oct
Demo	Round 2 video	1–2 min, shows steps 1–8 of §1	14 Oct 20:00
Demo	Finale reliability	5 of 5 rehearsals pass, ≥ 1 with internet off (GOAL-007)	23 Oct
Demo	Honest metrics slide	All 5 metrics + ≥ 2 failures (GOAL-009)	14 Oct

Working times in the preceding table are internal planning targets, not verified organiser requirements or delivery promises. Confirm the official cutoff and submission channel before setting the final schedule.

3 Non-goals

Category	Not doing
Features	Alerts of any kind; face matching; telecom location targeting; live police/CID data; SMS/IVR; native app; parent cross-device access; dashboard filters
Technical work deferred	Phone OTP for parents; migrations framework; background job queue; multi-region hosting; automated retention scheduler; LLM re-ranking; real officer verification
Production concerns	Load beyond 50 users; 24/7 uptime; backups beyond the managed host's defaults; formal legal review; security audit by a third party; data-sharing agreements

4 Rebased gates and capacity

The previous 9–14 October empty-repository assumption is historical. At issue time (11 October, Asia/Dhaka), actual completion and team availability are UNVERIFIED. Do not mark work done merely because a specification exists.

Gate	Target working window	Exit evidence
A Contract/inventory	11 Oct	Repo/deploy and task status; V2 fields/states/permissions approved; provider/audio/embedding evidence; official cutoff confirmed
B Complete typed journey	11–12 Oct	One synthetic case on one DB: manual/review/submit/output/found/rank/decision/referral record/close/counts/audit
C Quality and evaluation	12–13 Oct	Target PG16/role and concurrent tests; frozen manifest; live cache-disabled metrics and failures; device/a11y checks
D Record and submit	13–14 Oct	Tested script/topology, honest labels/timings/spend, export/upload/link checks, organiser submission evidence
E Finale hardening	15–24 Oct	Same-backend local rehearsal, five recorded runs including internet-off, backup recording, final test/limitations

If starting from zero on 11 October, four full days at 10 hours/person/day yield about 120 team-hours. Actual remaining work is the sum of unfinished estimates, not the full historic workload. The V2 added tasks below and dependency path are calculated in the handover. Estimates are planning inputs, not delivery promises. Team lead must reassign work/cut optional scope or report missing features if remaining work exceeds available hours. Preserve correctness, honest evaluation and submission buffer.

5 Task board

Owners: **S** = Sadman (lead/integration), **O** = Obidit (intake & LLM), **M** = Masnun (data, matching & dashboard). CP labels inherited from V1 are removed; computed V2 dependency path is in the handover. Owner capacity can change the actual schedule.

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-001	Repo: Next.js App Router, TS strict, Tailwind, ESLint, CI (lint, typecheck, test)	Must	S	—	3	CI green on `main`; preview deploy URL	
TASK-002	Pick DB host (OQ-021); apply `schema_v2.sql`; local Docker DB	Must	S	TASK-001	2	HNSW index created on host and locally	
TASK-003	Provider keys, `.env.example`, record real prices in `docs/costs.md` (OQ-004, OQ-022)	Must	O	—	1	Both LLMs and STT answer a test call	
TASK-004	Spike: local embedding model on Vercel (OQ-023, RISK-015)	Must	M	TASK-001	3	Report: cold start, size; decision local vs API	
TASK-005	Spike: Android MediaRecorder → STT (A-030)	Must	O	TASK-003	2	3 phone recordings transcribed	
TASK-006	Auth.js credentials, parent cookie, `requireRole`, seeded users	Must	S	TASK-002	5	TC-023 passes	
TASK-007	API layer: error envelope E-001 – E-024, Zod helpers, request ID, rate limiter (COMP-021)	Must	S	TASK-001	4	Unit tests for envelope and limiter	
TASK-008	Layout shell, `tokens_v2.css`, Noto Sans Bengali, language toggle (COMP-022)	Must	S	TASK-001	3	Bangla renders on 1 Android phone	
TASK-009	District data: 64 rows bn/en/centroids + neighbours (DEP-004, A-013)	Must	M	TASK-002	3	`npm run seed` loads 64 districts	
TASK-010	Synthetic set generator: ~200 records, ground truth, duplicates, decoys (A-039)	Must	M	TASK-009	6	Ground-truth file hashed and committed	
TASK-011	50 hand-labelled reports	Must	M	—	5	`eval/data/labelled.jsonl` reviewed by O	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-012	Normaliser: Bangla digits, time words, phone, district aliases, romaniser (COMP-008)	Must	O	**TASK-009**	5	30 unit tests pass	
TASK-013	LLM adapter: 2 providers, Zod contract, evidence check, retry, fallback (COMP-007)	Must	O	TASK-003, TASK-012	7	Extracts 10 sample texts; fallback test passes	
TASK-014	Speech adapter + API-001 (COMP-006)	Must	O	TASK-005, TASK-007	3	TC-001 steps 1–3 pass	
TASK-015	Case service, state guard, audit, versions (COMP-014, COMP-015)	Must	S	TASK-007, TASK-061, TASK-062	6	TC-032 and TC-016 pass	
TASK-016	Embedding service + `npm run reembed` (COMP-011)	Must	M	**TASK-004**	3	200 records embedded	
TASK-017	Matching engine: filters, cosine, bonuses, template reasons (COMP-012)	Must	M	TASK-015, TASK-016	7	TC-009 passes; first recall@5 printed	
TASK-018	Duplicate detector (COMP-013)	Must	M	**TASK-016**	3	TC-013 passes	
TASK-019	Seed loader with synthetic labels and 3 demo cases	Must	M	**TASK-010**	2	`npm run seed -- --set demo` < 2 min	
TASK-020	Intake orchestrator + API-002 + question templates (COMP-009)	Must	O	**TASK-013**	4	TC-003 passes	
TASK-021	SCREEN-001/002 intake chat + recorder (COMP-024 – COMP-027)	Must	O	TASK-008, TASK-014, TASK-020	7	Voice and text paths work on phone	
TASK-022	SCREEN-003 review + API-003 + draft persistence	Must	O	TASK-015, TASK-021	5	TC-004, TC-031 pass	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-023	Output templates GD/poster + wording guard + API-006 (COMP-010, COMP-016)	Must	O	**TASK-015**	4	TC-005, TC-024 pass	
TASK-024	SCREEN-004 outputs + poster PNG (COMP-030, COMP-031)	Must	S	**TASK-023**	3	TC-006 passes	
TASK-025	SCREEN-005 manual form fallback (COMP-032)	Must	S	**TASK-022**	2	TC-017 passes	
TASK-026	SCREEN-007 found-child log + API-007 (COMP-034)	Must	M	TASK-015, TASK-020	3	TC-008 passes	
TASK-027	SCREEN-009 queue + API-008 (COMP-035, COMP-036)	Must	M	**TASK-017**	3	Queue lists seeded candidates	
TASK-028	SCREEN-010 match review + API-009/010 (COMP-037, COMP-038)	Must	M	TASK-017, TASK-027	6	TC-010 (confirm part), TC-011 pass	
TASK-029	SCREEN-012 case detail, close, audit + API-011/013 (COMP-039, COMP-042)	Must	S	**TASK-028**	5	TC-010 full passes	
TASK-030	SCREEN-011 map + table + API-012 (COMP-040, COMP-041)	Must	M	TASK-009, TASK-029	5	TC-012 passes	
TASK-031	SCREEN-008 sign-in + SCREEN-016 (COMP-047, COMP-045)	Must	S	**TASK-006**	2	TC-022 passes	
TASK-032	SCREEN-013 admin: redact identifying payloads + verify [MOCK] + API-016/017 (COMP-043)	Must	S	**TASK-015**	3	TC-015 passes	
TASK-033	Demo-safe cache + fixtures for 3 cases + cached badge (COMP-017, COMP-048)	Must	O	TASK-020, TASK-019	3	TC-021 passes on laptop	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-034	Accessibility pass: axe + keyboard + 320 CSS px reflow, 200 % text zoom and 400 % viewport zoom	Must	S	TASK-030	3	TC-033 passes	
TASK-035	Bangla copy review; all empty/error/loading states	Must	O	TASK-030	3	Every E-xxx shows bn + en text	
TASK-036	Test on 2 Android phones (Bangla rendering, mic, 360 px)	Must	M	TASK-030	2	Checklist signed in `docs/device-test.md`	
TASK-037	Unit tests: normaliser, matcher, state guard, wording guard	Must	S	TASK-017	4	≥ 60 tests, CI green	
TASK-038	Playwright E2E: essential journey + safeguarding TCs	Must	S	TASK-030, TASK-032, TASK-059	5	TC-001–066 app/service checks where applicable; companion-only results kept separate	
TASK-039	Evaluation harness: smoke/dev run first; held-out run after manifest/model freeze	Must	M	TASK-011, TASK-017, TASK-018, TASK-064	5	TC-020 passes; report committed	
TASK-040	WER recordings (60 clips) + run	Must	O	TASK-014	3	WER in report	
TASK-041	Timing study, ≥ 5 testers, assistant vs manual form	Must	S	TASK-025	3	`eval/data/timing.csv`; median in report	
TASK-042	Production deploy + managed DB seed + smoke test	Must	S	TASK-038	2	`npm run smoke` passes on prod	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-043	Final evaluation run on freeze build	Must	M	TASK-042, TASK-039, TASK-040, TASK-041, TASK-060	2	Frozen cache-disabled report with all metrics, denominators, failures, commit/model/manifest hashes	
TASK-044	Round 2 deck, 5–8 slides	Must	S	**TASK-043**	4	Working 5–8-slide deck; exact organiser deliverables confirmed; honest measured/ pending labels	
TASK-045	Round 2 demo video, 1–2 min	Must	O	**TASK-042**	4	Working 1–2-minute MP4; confirm official format; compressed/cached timings labelled	
TASK-046	README, Drive folder (links TXT, deck, video), reply to selection email	Must	S	TASK-044, TASK-045	1	Receipt via organiser-confirmed channel before confirmed cutoff; links accessible	
TASK-047	13219 summary (FR-010)	Should	O	**TASK-023**	1	TC-007 passes	
TASK-048	Synthetic fixture import + API-015 (FR-024)	Should	M	**TASK-020**	3	TC-014 passes	
TASK-049	Edit after submit + API-005 + SCREEN-006 (FR-029)	Should	O	**TASK-022**	4	TC-018 passes	
TASK-050	Parent status API-018 (FR-030)	Should	O	**TASK-049**	1	TC-019 passes	
TASK-051	Duplicate review SCREEN-014 (FR-023)	Could	M	**TASK-018**	3	Mark same/different works	
TASK-052	Disaster mode SCREEN-015 (FR-036)	Could	M	**TASK-026**	2	TC-035 passes	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-053	Finale local stack: Docker DB, model cache, fixtures, `DEMO_MODE=1`	Must	S	**TASK-033**	4	Laptop runs journey offline	
TASK-054	5 rehearsals, ≥ 1 internet-off	Must	S, O, M	**TASK-053**	5	Rehearsal log 5/5	
TASK-055	Finale deck, 5–10 min	Must	S	**TASK-043**	4	Timed run ≤ 10 min	
TASK-056	Fixes from mentor session 2 (12–13 Oct)	Should	S, O, M	—	5	Agreed fixes merged by 14 Oct 18:00	
TASK-057	5–10 user conversations, only if OQ-019 allows	Should	S	—	4	Notes with consent; findings slide	
TASK-058	Backup recording of the finale demo	Must	O	**TASK-054**	2	MP4 on laptop and Drive	
TASK-059	`npm run spend` + spend cap check (DECISION-017)	Must	O	TASK-013, TASK-063	1	TC-036 passes	
TASK-060	Performance check against D2 Part B §12.1	Must	M	**TASK-042**	2	TC-034 results in report	
TASK-061	Apply V2 required fields, DTOs, token/ idempotency and referral/count semantics	Must	S	**TASK-001**	3	Runtime schemas/examples and role projections agree; FV-01–05/10 closed	
TASK-062	V2 schema invariants, restricted roles and full related-payload redaction	Must	S	TASK-002, TASK-061	4	Negative and positive lifecycle/redaction checks; target PG16 CI pending until run	
TASK-063	Atomic paid-call counters/reservations, replay handling and price sheet	Must	O	TASK-003, TASK-062	2	Draft replay/concurrency/ budget checks; manual path survives paid refusal	

ID	Task	Priority	Owner	Dependencies	Hours	Completion evidence	Plan note
TASK-064	Identity-grouped manifest, no-match/ duplicate hard negatives, frozen metrics	Must	M	TASK-010	3	Manifest counts/split/hash asserted before held-out evaluation	

Critical path: The V1 printed chain was invalid (67 hours, not ~59; four undeclared links). V2 dependency/resource summary is generated from the task table in TEAM_HANDOVER_V2. A dependency longest path is a lower bound, not a resource-feasible finish date.

6 Folder structure

```
fera/
├─ app/                          # Next.js App Router
│  ├─ (parent)/                  # SCREEN-001–006, no auth
│  │  ├─ page.tsx                # SCREEN-001 welcome
│  │  ├─ report/page.tsx         # SCREEN-002 intake chat
│  │  ├─ review/page.tsx         # SCREEN-003
│  │  ├─ case/[ref]/page.tsx     # SCREEN-004 / SCREEN-006
│  │  └─ form/page.tsx           # SCREEN-005 manual form
│  ├─ staff/                     # SCREEN-007, SCREEN-015
│  ├─ officer/                   # SCREEN-009–012, SCREEN-014
│  ├─ admin/                     # SCREEN-013
│  ├─ signin/page.tsx            # SCREEN-008
│  ├─ not-found.tsx              # SCREEN-016
│  └─ api/                       # API-001 – API-020 route handlers
├─ components/                   # COMP-022 – COMP-050 (UI)
├─ lib/
│  ├─ ai/                        # COMP-006 speech, COMP-007 llm, COMP-011 embeddings, providers/
│  ├─ intake/                    # COMP-008 normaliser, COMP-009 orchestrator, question templates
│  ├─ outputs/                   # COMP-010 templates, COMP-016 wording guard
│  ├─ cases/                     # COMP-014 case service, COMP-015 audit
│  ├─ matching/                  # COMP-012 matcher, COMP-013 dedupe
│  ├─ auth/                      # COMP-005
```

```
│   ├─ http/                    # COMP-004 envelope, COMP-021 rate limiter, errors E-001–E-024
│   └─ db.ts                    # pg pool
├─ fixtures/demo/               # COMP-017 cached transcripts/extractions/embeddings
├─ scripts/                     # seed, reembed, model:fetch, spend, smoke
├─ eval/                        # COMP-020 metric scripts, data/, report-YYYY-MM-DD.md
├─ tests/                       # vitest unit + playwright e2e
├─ styles/tokens.css            # D4 §16
├─ schema.sql                   # D2 §7
├─ docs/                        # D1–D7, costs.md, device-test.md
├─ docker-compose.yml           # local Postgres + pgvector
└─ .env.example
```

7 Testing strategy

Level	Tool	Scope	When
Unit	Vitest	Normaliser, evidence check, matcher scoring, dedupe rule, state guard, wording guard, templates	Every push (CI)
Integration	Vitest + local Postgres	API handlers against a seeded DB; trigger and constraint behaviour	Every push
End-to-end	Playwright (mobile 360 px + desktop 1280 px)	Essential journey; safeguarding TCs; error flows	Nightly + before freeze
Accessibility	@axe-core/playwright + manual keyboard and screen-reader spot check	All Must screens, light and dark	13 Oct, 23 Oct
Manual demo checks	Checklist on 2 Android phones + laptop	Bangla rendering, mic, copy, PNG, map	13 Oct, each rehearsal
Failure-state tests	Playwright with network throttling/offline; provider keys removed	E-010 – E-014, E-022 – E-024, fallback form, demo cache	13 Oct
AI evaluation	`/eval` scripts	matching/duplicate/extraction/speech metrics, abstention/coverage, timing and audit coverage	11 Oct (first), 13 Oct, 14 Oct (final), 23 Oct

8 Test cases

TC	Covers	Preconditions	Steps	Expected result	Pass if
TC-001	STORY-001, FR-002	Phone, mic allowed, preview deploy	1 Open SCREEN-002. 2 Record 38 s scripted Rafi note. 3 Stop.	Editable Bangla transcript appears	Shown ≤ 20 s; contains "রাফি" and "১৩" or "13"
TC-002	STORY-002, FR-001, FR-003, FR-004, FR-033	—	Submit "আমার ছেলে রাফি, বয়স ১৩, গতকাল বিকেলে মাদ্রাসা থেকে ফেরেনি। নম্বর ০১৭১১২২৩৩৪৪"	Draft: age 13, sex male, explicit fixture-clock yesterday afternoon interval with 16:00 display point, phone 01711223344; district empty	Required fields and supported time bounds correct; unknowns explicit; draft ≤ 15 s; UI in Bangla
TC-003	STORY-003, FR-005	TC-002 draft	Read question; answer "মিরপুর, ঢাকা"	One question about district; then draft complete	Exactly 1 question per missing field; district code = Dhaka
TC-004	STORY-004, FR-006, FR-007, BR-004	Complete draft	1 Query DB and officer queue before submit. 2 Edit clothing. 3 Tick check. 4 Submit.	Before submit: 0 child case rows; metadata allowed. After: case open with edited clothing	0 child case rows pre-submit; metadata/ cost rows allowed; stored = edited values
TC-005	STORY-005, FR-008, BR-022	Submitted case	Tap "জিডি খসড়া"	GD text with name, age, place, time, phone; "খসড়া" label	All 5 fields present; no token absent from the case fields except template text
TC-006	STORY-006, FR-009, BR-006	Submitted case	Tap "পোস্টার"; download PNG	Text + PNG with 13219 and phone; no photo	PNG renders Bangla; no image input exists
TC-007	STORY-007, FR-010	Submitted case	Tap "১৩২১৯ সারাংশ"	Bangla summary	≤ 80 words
TC-008	STORY-008, FR-011	Staff signed in	Log Kamalapur found boy (demo data)	Ref F-... shown; "মিল খোঁজা হয়েছে"	Case open or matched_pending_review ≤ 15 s; no candidate details shown to staff
TC-009	STORY-009, FR-012 – FR-014, BR-017, BR-018	Seeded set + TC-008	Officer opens queue	Rafi case with ≤ 5 candidates, scores, reasons	Scripted fixture true pair rank recorded; held-out quality reported separately; no candidate violates eligibility (SQL check)
TC-010	STORY-010, STORY-012, FR-015, FR-016, FR-018, BR-001, BR-005	**TC-009**	1 Open candidate. 2 Confirm (2 steps). 3 Close with "সমন্বয়ের সিদ্ধান্ত ও রেফারেল নথিভুক্ত(সিমুলেশন)".	Missing confirmed → closed; found referred; 1 referral; audit rows	All states correct; referral exists; 3 audit entries

TC	Covers	Preconditions	Steps	Expected result	Pass if
TC-011	STORY-011, FR-017, BR-019	Candidate pending	Reject with reason "পোশাক আলাদা"; re-run API-014	Candidate rejected; case open; pair absent after re-run	Pair not re-suggested
TC-012	STORY-013, FR-019, BR-014	Seeded + TC-010	Open map and table	Counts per district	Every district count = SQL count by group
TC-013	STORY-014, FR-022, FR-023	Mim parent case + public duplicate	Open either case	Banner "সম্ভাব্য একই শিশু"; both flagged reports remain independently visible	Link exists; distinct-child shared-phone controls stay visible
TC-014	STORY-015, FR-024	Admin	Import 5 invented synthetic sample texts with example.invalid URLs, one containing "ignore previous instructions"	5 synthetic cases tagged simulated public_import with invented URLs	Injected text yields only schema fields; no server fetch of URLs
TC-015	STORY-016, FR-025, BR-009	3 closed/referred, 1 open	Run redact identifying payloads all eligible	eligible payloads redacted, active linked evidence skipped	direct and related identifying payloads cleared; pseudonymous counts unchanged
TC-016	STORY-017, FR-021, BR-010, NFR-017	TC-010 done	1 Read audit. 2 Try SQL UPDATE/ DELETE on audit_log	Entries for confirm, referral, close	All present; UPDATE/DELETE/TRUNCATE raise error; runtime cannot alter guards
TC-017	STORY-018, FR-026	LLM keys invalidated	Submit text	Manual form opens pre-filled with any parsed fields	Form ≤ 2 s after error; text kept
TC-018	STORY-019, FR-029, BR-025	Case open with pending candidate	Edit age; confirm	Version 2; candidates voided; re-match run	Version row exists; confirmed case returns E-018
TC-019	STORY-020, FR-030, BR-007	Parent case matched	Parent opens status and own scoped edit read	"যাচাই চলছে" plus own fields if editable	Response JSON has no found-case or candidate fields
TC-020	STORY-022, FR-028, GOAL-002 – GOAL-005, GOAL-009	Labelled sets	`npm run eval:all`	Report with 5 metrics, sizes, SHA, ≥ 2 failures	All sections present

TC	Covers	Preconditions	Steps	Expected result	Pass if
TC-021	STORY-023, FR-027, FR-032, NFR-007	`DEMO_MODE=1`, laptop offline	Run Rafi journey with scripted inputs	Journey completes; "ক্যাশ" badges; "সিনথেটিক" labels	Completes with network off
TC-022	BR-007, FR-020	Parent and staff sessions	Request API-009 and API-004 for another case	404 body	E-004 returned; `access.denied` audit row
TC-023	BR-001, FR-035, FR-037	Staff and unverified officer	POST API-010 confirm	Refused	E-003/E-004; candidate still pending
TC-024	BR-002, BR-003, FR-034	Template with "পালিয়ে" injected in test	Generate outputs	Blocked	E-016; text not shown
TC-025	BR-002	Submitted case	Inspect all outputs and UI	Draft labels; no "send alert" control	Every output labelled "খসড়া"; 0 alert actions in UI
TC-026	BR-006	—	Search UI and API schema for image inputs	None	0 file inputs except audio
TC-027	BR-008	—	Compare `cases` columns and forms with D1 BR-015 + optional list	Match	No extra personal fields
TC-028	NFR-010	Full E2E run	Regex scan logs for `01[0-9]{9}` and seeded names	None found	0 matches
TC-029	BR-020	Found case with 2 pending candidates	Confirm one	Other superseded; second confirm on same found fails	Unique index blocks second confirm
TC-030	BR-015, BR-016	—	Submit age 19, future time, phone 0171, district "XYZ"	4 field errors	E-001 with 4 paths
TC-031	FR-031	Submitted case	Concurrently replay the same submission UUID and canonical payload	Existing ref shown	200 stable retry reference; 1 row; changed payload conflicts

TC	Covers	Preconditions	Steps	Expected result	Pass if
TC-032	BR-012, BR-013	Seeded cases	SQL: open→confirmed; found→closed	Rejected by trigger	Both raise `check_violation`
TC-033	NFR-011	Prod build	axe on Must screens, light and dark; keyboard walk	No serious/critical	0 serious/critical; all actions keyboard-reachable
TC-034	NFR-001 – NFR-005	Prod build	Lighthouse mobile; `eval/perf.ts`	Values recorded	Each target met or reported as missed
TC-035	STORY-021, FR-036	Staff	Open disaster screen; log child	Banner "পরিকল্পিত সম্প্রসারণ"	Saved as found case
TC-036	NFR-016, DECISION-017	Spend cap set to current total	Call API-002	Refused	E-023; manual form opens
TC-037	BR-011	Deck, UI, README	Search for compliance claims	None	0 unsupported compliance claims; cited legal facts have primary sources

8.1 V2 additional acceptance cases

TC	Case	Covers	Pass condition
TC-038	Shared-phone distinct children	BR-021, FV-02	Two siblings/shelter reports remain searchable; phone-only link cannot confirm_same
TC-039	Unknown found age	BR-017, FV-03	Eligible with unknown cue; no fabricated age agreement
TC-040	Overlapping time uncertainty	BR-016/017	Interval-compatible found report is not removed by midpoint
TC-041	No plausible candidate	BR-018	Qualified shortlist may be empty; no misleading confidence
TC-042	Concurrent submission UUID	FR-031, FV-05	One case; stable response; different payload conflicts
TC-043	Draft replay/concurrency	BR-023, FV-05	Server usage cannot reset; six dispatched calls maximum
TC-044	Admin rematch projection	BR-007/024	Status only, no candidate IDs/scores/child payload

TC	Case	Covers	Pass condition
TC-045	Owner editable read	API-004, SCREEN-006	Own fields before lock; no other-party payload
TC-046	Stale candidate decision	**API-010**	Expected-version conflict, no new decision/referral
TC-047	Wrong referral endpoints	FV-06	Constraint rejects mismatched candidate pair
TC-048	Reject without reason	**BR-001**	Service and DB reject NULL/short reason
TC-049	Terminal report edit	**BR-025**	Blocked; privileged redaction separate; no silent history rewrite
TC-050	Related note redaction	**BR-009**	Outcome/referral/decision/candidate/provenance/output payloads cleared; IDs remain pseudonymous
TC-051	Audit truncate/privilege	**BR-010**	UPDATE/DELETE/TRUNCATE rejected; runtime no DDL/ownership
TC-052	Unknown-resource denial	**API-004**	Safe audit recorded with null FK and no information leak
TC-053	Last competitor invalidated	**BR-020**	Other affected report returns to open if no pending candidate
TC-054	Matching fails after submit	API-003/014	Saved report, explicit failed/queued state, reliable retry
TC-055	One pair dashboard lifecycle	**BR-014**	Missing/found categories change separately; pair count one
TC-056	Evaluation fixture contamination	**GOAL-009**	Quality run rejects/excludes cache/fixture hits and labels result
TC-057	Single-DB offline scenario	FV-08	All local roles see same new report; assets preloaded
TC-058	Reflow/keyboard/Bangla	**NFR-011**	320px/400%/200% checks; shortcuts disabled during typing and switchable
TC-059	Embedding space switch	FV-09/15	No mixed vectors; matching paused until validated rebuild
TC-060	Provider notice/metadata	BR-004, FV-12	Notice before dispatch; no child payload in pre-submit DB metadata
TC-061	Encoded audio limits	**BR-023**	Real phone codec/duration/size tested; WAV >2 MB rejected or compressed
TC-062	PostgreSQL16/runtime host	D2B acceptance	Original V2 SQL applies and meaningful constraints tested on target PG16/roles; portable engine result alone not enough
TC-063	Correction/reopen pilot gate	**BR-025**	Documented two distinct verified reviewers, reason/evidence/history; real data blocked until implemented/approved
TC-064	Manifest/group split	FV-11	200 count, identities/no-match/duplicates declared; no connected-group leakage

TC	Case	Covers	Pass condition
TC-065	Concurrent AI budget	FV-12/13	Reservations cannot exceed cap; uncertain usage retained; duplicate settlement idempotent
TC-066	Legal claims/source accuracy	BR-011	No unsupported compliance or integration claim; accurate primary-law citations allowed

9 Demo plan

9.1 Setup

Item	Detail
Devices	Online: all roles on deployed origin/one managed DB. Local: all roles on laptop stack/one local DB, in separate browser profiles; phone only with tested secure LAN access. Spare laptop preloaded.
Build	Round 2: production deploy. Finale: production deploy + laptop-local copy with `DEMO_MODE=1`
Accounts	`officer.karim` (verified), `staff.rumana`, `admin` — passwords in the team vault
Reset	`npm run seed -- --reset --set demo` 10 min before each run
Checks	`/api/health` green; `npm run spend` < cap; phone font check

9.2 Demo data (synthetic, labelled "সিনথেটিক")

Name	Record	Role in demo
রাফি (Rafi)	Missing boy, 13, blue shirt, last seen 8 Oct afternoon interval, display reference 16:00; fixture intake clock explicitly 9 Oct near madrasa, Mirpur, Dhaka	Live parent voice report
Kamalapur boy	Found boy, est. 12–14, blue shirt, scar on left hand, Kamalapur railway station, Dhaka, 9 Oct 07:30	Logged live by staff; true match
Narayanganj decoy	Found boy, 13, red shirt, Narayanganj	Rank 2–3; rejected in finale version
মীম (Mim)	Missing girl, 11, Tongi, Gazipur; parent report + public-post duplicate	Shows duplicate banner
সাব্বির (Sabbir)	Missing boy, 15, Chattogram; no found record	Shows open case and empty candidates
~195 others	Seeded synthetic set	Fills map

9.3 Actions and expected results

#	Action	Expected on screen
1	Phone: SCREEN-001 → Start	Intake chat in Bangla

#	Action	Expected on screen
2	Record the Rafi script (38 s)	Transcript appears
3	Send; answer district "মিরপুর, ঢাকা"	Review cards filled
4	Tick check; submit	Ref M-...; status "খোলা"
5	Tap GD draft and poster	Bangla GD text; poster preview
6	Laptop: staff logs Kamalapur boy	Ref F-...; "মিল খোঁজা হয়েছে"
7	Officer: queue	Rafi at top, score shown
8	Open; side-by-side; confirm (2 steps)	"পর্যালোচকের সিদ্ধান্ত... রেফারেল নথিভুক্ত (সিমুলেশন)"
9	Close case	"কেস বন্ধ হয়েছে"
10	Map	Dhaka missing reports: Closed +1, Matched −1; found referral-recorded unchanged; confirmed pairs unchanged
Finale extras	Reject Narayanganj decoy with reason; show Mim duplicate banner; show audit trail; show metrics slide	As in D3

9.4 Backup paths

Failure	Backup
Speech provider slow/ down	Type the scripted Rafi text; or `DEMO_MODE=1` scripted note → cached transcript (badge "ক্যাশ")
LLM down	Fallback provider; then cached extraction; then manual form
Venue internet down	Restart the whole labelled scenario on the local stack, with all roles sharing its DB. Never mix a new cloud report with local officer state. If local fails, play labelled backup recording (TASK-058).
Laptop failure	Spare laptop with same local setup
Map tiles fail	Bundled district outlines + table; no bulk offline caching of public OSM tiles

9.5 Versions

Version	Length	Content
Round 2 video	1:45	Steps 1–10, sped-up waits, English captions over Bangla UI
Finale live	7–8 min of a 10 min slot	Problem (1 min) → live steps 1–10 (4 min) → extras (1.5 min) → metrics, safeguards, scale (1.5 min)

Spoken scripts and timings are in D7 §3.

10 Hackathon checklist

Area	Item	Done
Product	Essential journey on prod (TC-001 – TC-012)	☐
Product	Should items only if Must is green	☐
UI	Bangla default, English toggle, synthetic/cached/draft labels	☐
Backend	State trigger, audit append-only, spend cap	☐
Security	`npm audit` clean; no secrets in repo; rate limits; E-004 for hidden records	☐
Accessibility	TC-033 green, light and dark	☐
Testing	Unit + E2E green; failure-state tests run	☐
Evaluation	Report with 5 metrics + failures, dated 14 Oct	☐
Deployment	Prod smoke test; `r2-freeze` tag	☐
Presentation	5–8 slides incl. metrics slide and [MOCK] items marked	☐
Submission	GitHub repo link	☐
Submission	5–8 slide PowerPoint deck	☐
Submission	1–2 min demo video	☐
Submission	Drive folder with TXT/DOC of links + deck + video	☐
Submission	Reply to selection email with folder link before 14 Oct; exact cutoff pending organiser confirmation	☐
Finale	Local stack, 5 rehearsals, backup recording, spare laptop	☐

11 Recommendation

Topic	Advice
Build first	Critical path in §5: case service + state guard, extraction, intake screens, found log, matcher, match review, close, map. Text path before voice.
Cut if short (in order)	1 FR-036 disaster screen → 2 FR-010 13219 summary → 3 FR-024 import → 4 FR-029 edit after submit → 5 live voice (keep cached scripted note, label it) → 6 poster PNG (keep poster text). Never cut review-before-submit, officer confirm, referral, audit or redact identifying payloads.
Impact (judges)	Show one child's case reaching a known status, and the closure count on the map; tie to "2,038 still untraced" (pitch deck).
Feasibility	Show measured metrics with failures, the USD spend so far, and the offline demo.

Topic	Advice
Scalability	Show 64-district map, 30,000-cases/year design (NFR-015), provider adapter, and that CID integration is a data agreement, not new infrastructure.
Do not attempt	Face matching, alert sending, scraping social media, real child data, real officer verification, a native app.

Assumptions (D6)

ID	Assumption
A-040	Actual availability and completed work must be inventoried on 11 Oct. A fresh 11–14 Oct start at 10 h/person/day gives at most 120 team-hours; this is an assumption, not verified capacity.

Open questions (D6)

ID	Question	Default
OQ-024	Do classes, exams or labs reduce team hours before 14 Oct?	UNKNOWN; inventory actual availability and finished work at the 11 Oct review

Risks (D6)

ID	Risk	L	I	Mitigation	Owner
RISK-017	V2 Round 2 estimates total 184 h and may exceed actual remaining capacity	M	H	Daily stand-up; inventory completed work and actual owner capacity, sequence dependencies, reassign and cut optional scope on 11 Oct	Sadman

IDs introduced

- TASK-001 – TASK-060 (one line each in §5)
- A-040 Team capacity
- OQ-024 Team availability
- RISK-017 Capacity overrun

IDs referenced

- FR-001 – FR-037, STORY-001 – STORY-023, BR-001 – BR-022, BR-025, NFR-001 – NFR-017 (D1)
- TC-001 – TC-037 (D1, written here)
- GOAL-001 – GOAL-009 (00)
- SCREEN-001 – SCREEN-016 (D1)
- API-001 – API-020, COMP-004 – COMP-021, E-001 – E-024, DECISION-017, A-013, A-030, A-039 (D2)
- COMP-022 – COMP-050 (D4)
- OQ-004, OQ-019, OQ-021 – OQ-023, RISK-015 (00, D1, D2)
- DEP-004 (D1)

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